

AMLESH SWARUP SAHOO

Electrical Engineering Student | Embedded Firmware • PCB Design • Hardware-Software Integration

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EDUCATION

BITS Pilani – Iowa State University (ISU) 2+2 Dual-Degree Program

2024 – 2028

B.E. Electronics and Communication Engineering (BITS Pilani) + B.S. Electrical Engineering (Iowa State University)

First two years at BITS Pilani, Pilani Campus; final two years at Iowa State University, Ames, IA

Relevant Laboratory Coursework: Digital Design Lab; Analog Electronics Lab

TECHNICAL SKILLS

Firmware & Programming: C, C++, Embedded C, Git/GitHub

Microcontrollers & Peripherals: STM32H7, ESP32, STM32CubeIDE, Arduino IDE; ADC, DMA, Timers, GPIO, UART, SPI, I2C

Connectivity & IoT: WebSocket, Bluetooth Low Energy (BLE), Bluetooth Classic, Ethernet (W5500), JSON Data Serialization

PCB & Circuit Design: KiCad (2-layer/4-layer PCB), LTspice, Cadence, Tinkercad, AutoCAD; Power, Sensing & Buck-Boost Circuits

Hardware Test & Debug: Oscilloscope, Digital Multimeter, Function Generator, Bench DC Power Supply, Soldering, Prototyping

Application & Data Tools: Flutter/Dart, fl_chart, CustomPainter, CSV Data Export

EXPERIENCE

Undergraduate Research Intern — Iowa State University, Prof. Manojit Pramanik

2026 – Present

Project: Development of an Embedded Processing System for Pulsed-LED Photoacoustic Tomography Using a Microcontroller Platform

- Architecting an STM32H7-based real-time acquisition platform for pulsed-LED photoacoustic tomography, integrating deterministic sampling, memory buffering, embedded processing, and system-level hardware-software coordination.
- Developing the acquisition chain around timer-triggered ADC, DMA-to-RAM buffering, peripheral timing, and high-speed STM32-to-ESP32 communication while evaluating throughput, memory, latency, and interface tradeoffs for reliable data transfer.

Summer Research Intern — Indian Institute of Technology (IIT) Hyderabad, Prof. Rupesh Wandhere June 2025 – July 2025

- Developed and validated ESP32-based embedded/IoT systems spanning real-time power monitoring, touchscreen interfaces, Ethernet/WebSocket communication, Bluetooth connectivity, and Flutter-based visualization/control.
- Integrated ST7789 and ILI9341/XPT2046 displays, W5500 Ethernet, BLE/Bluetooth Classic, voltage/current sensing, and buck-boost hardware; streamed JSON telemetry and live waveforms to Flutter applications with CSV recording/export.
- Scaled monitoring from single-device prototypes to 4 ESP32 nodes with parallel WebSocket channels and priority-based message filtering, while improving reliability through selective redraw, circular buffering, exponential smoothing, touchscreen calibration, and reconnect/loss-recovery logic.

PROJECTS

MCU Data Logger PCB Design — 2-Layer & 4-Layer

- Developing a microcontroller-based data-acquisition board that captures analog and sensor measurements, stores logged data, and transfers measurements to a host system for monitoring and analysis.
- Designed in both 2-layer and 4-layer KiCad variants integrating the MCU, regulated power and decoupling, data-storage interface, sensor/analog inputs, communication interfaces, and programming/debug headers.

Programmable Power Supply (PPS) & Photovoltaic (PV) Monitoring System

- Engineered an ESP32 monitoring system that measures 6 real-time parameters—voltage, current, power, temperature, duty cycle, and efficiency—for programmable power supplies and PV sources using current-sensing and buck-boost circuitry.
- Built a 3-mode communication layer (WebSocket/W5500, Bluetooth Classic, BLE) and Flutter interface with Vset/Iset control, live fl_chart visualization, CSV export, and reconnect/loss-recovery logic.

Priority-Based WebSocket Messaging System for Multi-Node ESP32 Networks

- Designed a real-time priority-arbitration protocol across 4 ESP32 boards on W5500 Ethernet modules, where a central receiver accepts JSON data (parsed via ArduinoJson) only from the highest-priority of 3 sender nodes.
- Validated priority handling across all 3 sender nodes through testing, demonstrating deterministic accept/reject behavior for congestion control in distributed monitoring networks.

ACHIEVEMENTS & LEADERSHIP

- Selected Member, Student Technical Team (SUTT) — App Development Vertical, BITS Pilani
- Finalist, IEEE Electrical Systems Vertical Competition
- Member, BITS Pilani Institute Badminton Team — competed at the institute level